

BENI-MELLAL, MoroccoWMO: **601910**Lat: **32.367N**Lon: **6.400W**Elev: **1549**StdP: **13.89**Time Zone: **1.00 (E01)**Period: **94-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.3	36.5	28.0	23.2	56.0	30.8	26.4	51.5	14.7	52.2	12.6	52.3	2.5	120	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.2	106.3	68.6	102.9	68.7	99.3	68.6	73.0	94.9	71.8	93.4	70.8	92.1	7.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	103.5	81.3	65.2	98.8	80.0	64.0	94.6	79.0	37.6	94.9	36.5	93.6	35.5	92.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	11.6	9.7	31.3	110.2	3.1	2.3	29.0	111.9	27.2	113.2	25.4	114.5	23.1	116.2		
(5)			29.0	75.4	3.5	1.9	26.5	76.8	24.4	77.9	22.5	78.9	19.9	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	51.3	54.6	59.3	63.5	69.0	77.5	83.6	82.7	76.0	69.2	59.2	53.4		
	DBStd	12.14	4.21	4.72	5.38	5.04	6.69	5.49	5.68	4.67	5.32	4.96	5.82	4.19		
	HDD50	67	34	13	3	0	0	0	0	0	0	0	2	15		
	HDD65	1592	426	291	189	85	33	0	0	0	1	16	192	359		
	CDD50	6157	74	142	291	406	590	826	1042	1013	779	595	277	122		
	CDD65	2204	0	0	12	41	158	376	577	548	329	146	17	0		
	CDH74	28488	3	42	362	764	1986	4612	8092	7227	3561	1556	282	1		
	CDH80	15550	0	2	71	262	958	2534	5075	4287	1737	572	52	0		
(14) Wind	WSAvg	4.4	3.0	3.6	3.8	4.5	5.0	5.4	5.7	5.7	4.8	4.0	3.4	3.4		
(15) Precipitation	PrecAvg	13.9	1.7	1.6	1.7	1.7	0.9	0.4	0.2	0.1	0.5	1.3	1.8	2.0		
	PrecMax	24.6	6.1	4.5	6.4	6.0	3.3	2.6	0.8	0.5	2.9	4.6	7.1	5.6		
	PrecMin	6.9	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0	0.0		
	PrecStd	4.3	1.6	1.2	1.5	1.5	1.0	0.6	0.2	0.2	0.7	1.1	1.9	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	78.8	86.2	92.4	102.1	107.8	111.1	108.8	103.4	93.8	85.4	72.7		
		MCWB	54.9	57.4	60.6	63.0	65.5	68.1	68.5	68.3	67.9	64.0	60.8	56.3		
	2%	DB	69.2	74.6	81.6	87.3	95.2	102.5	107.7	104.5	97.6	89.4	80.9	69.8		
		MCWB	54.9	56.4	60.1	61.7	65.3	67.7	68.5	69.4	67.6	63.5	59.8	56.1		
	5%	DB	66.5	71.8	78.1	82.4	90.5	97.9	104.0	100.8	93.6	86.1	77.4	67.3		
		MCWB	53.9	56.6	58.9	61.0	65.0	67.8	69.1	69.4	67.2	63.1	59.2	55.7		
	10%	DB	63.4	68.3	74.4	78.1	84.8	93.4	99.7	97.2	89.5	82.8	72.7	64.6		
		MCWB	53.2	55.6	58.2	60.1	64.1	67.2	68.9	69.5	66.7	63.2	58.2	54.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	62.0	65.4	66.8	70.3	72.7	75.7	74.9	72.3	68.1	64.6	62.3		
		MCDB	66.8	71.1	79.2	84.6	89.6	94.5	99.8	95.9	90.9	83.5	78.0	66.9		
	2%	WB	57.2	59.9	62.9	64.3	67.6	71.1	73.2	73.0	70.7	66.7	62.2	59.5		
		MCDB	64.9	68.8	76.3	81.2	87.7	92.0	95.9	94.6	88.8	81.1	74.9	64.5		
	5%	WB	55.3	58.0	61.0	62.6	66.0	69.8	71.9	71.8	69.3	65.6	60.6	57.6		
		MCDB	63.6	67.7	73.5	77.8	85.8	91.3	94.5	93.5	87.2	79.2	73.0	64.0		
	10%	WB	53.6	56.2	59.3	61.0	64.3	68.5	70.6	70.7	68.2	64.4	59.0	55.9		
		MCDB	61.8	66.2	70.8	75.4	82.5	88.7	93.0	92.1	85.1	77.6	70.5	62.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.5	23.8	24.7	24.2	24.9	27.5	29.2	27.6	23.7	24.2	24.4	22.5		
		MCDBR	29.8	32.0	32.9	34.2	34.6	34.9	33.7	32.5	30.4	31.0	31.5	28.0		
		MCWBR	16.8	16.4	14.5	13.0	10.8	9.4	8.3	8.6	9.0	10.9	13.4	15.4		
	5% WB	MCDBR	24.8	25.8	28.2	29.2	30.0	30.2	29.3	28.6	25.1	23.9	26.1	21.0		
		MCWBR	15.2	14.6	14.3	12.3	10.5	9.5	8.9	8.8	8.3	9.4	11.9	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.335	0.394	0.410	0.451	0.495	0.562	0.538	0.453	0.408	0.339	0.303		
	taud		2.541	2.405	2.216	2.201	2.094	1.977	1.806	1.879	2.116	2.242	2.456	2.535		
	Ebn at Noon		298	292	280	279	268	255	237	241	258	262	276	287		
	Edh at Noon		26	33	43	46	51	58	68	62	47	39	29	25		
(44) All-Sky Solar Radiation	RadAvg		1093	1367	1730	2038	2251	2481	2423	2197	1858	1453	1129	1001		
	RadStd		94	126	106	125	142	110	68	81	92	75	83	76		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.43	N/A	-2.21	+1.57	-0.97	-0.83	N/A	N/A	N/A	N/A	+153

Nomenclature: See separate page